Lecture 5 Exam Scribe

L5_A_S2.pdf
PDF

generate a lecture scribe suitable for exam preparation from the provided lecture 5. While making this, reason step by step using definitions, assumptions, derivation, proof, and example solution exactly as given in the slides. Include the step by step reasoning so that when we revise for our exams it becomes easy. All explanations and reasoning must be from the provided ppt. do not introduce any new examples or theory apart from what is there in the ppt. Explain every definition and assumption explicitly. derive the formulas and solve the examples step by step using whatever is there in the slide. The tone should be formal and academically suitable for revision

Absolutely VIRJAAAA 💪📚
Here is a **formal, exam-revision lecture scribe** for **Lecture 5** created **ONLY from your provided PPT** and following the exact definitions, assumptions, derivations, proofs, and examples shown in the slides.

CSE400 — Lecture 5 Scribe (Exam Revision Notes)

Bayes' Theorem, Random Variables, and Probability Mass Function (PMF)

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1) Bayes' Theorem

1.1 Weighted Average of Conditional Probabilities

Step 1: Event Decomposition (Core Idea)
Let **A** and **B** be events.

We may express event **A** as:

$$A = AB \cup AB^c$$

Reasoning (as stated in the slides):

For an outcome to be in **A**, it must either:

- be in both **A and B**, or
- be in **A but not in B**.

Step 2: Mutual Exclusiveness

The events AB and AB^c are **mutually exclusive** (cannot occur together).

Step 3: Apply Axiom 3 (Additivity for Disjoint Events)

By Axiom 3:

$$Pr(A) = Pr(AB) + Pr(AB^c)$$

Step 4: Convert Intersections into Conditional Probability Form

Using:

$$Pr(AB) = Pr(A \mid B)Pr(B)$$

and similarly,

$$Pr(AB^c) = Pr(A \mid B^c)Pr(B^c)$$

So:

$$Pr(A) = Pr(A \mid B)Pr(B) + Pr(A \mid B^c)Pr(B^c)$$

Since:

$$Pr(B^c) = 1 - Pr(B)$$

Final form:

$$Pr(A) = Pr(A \mid B)Pr(B) + Pr(A \mid B^c)[1 - Pr(B)]$$

Interpretation (Exactly from slides)

> The probability of event A is a **weighted average** of the conditional probabilities, with weights given by the probability of the event on which it is conditioned has of occurring.

1.2 Learning by Example — Example #3.1 (Part 1/2)

Problem Statement

An insurance company divides people into:

- accident prone
- not accident prone

Given:

- $Pr(\text{accident in 1 year} \mid \text{accident prone}) = 0.4$
- $Pr(\text{accident in 1 year} \mid \text{not accident prone}) = 0.2$
- $Pr(\text{accident prone}) = 0.3$

Find:

Probability a new policyholder has an accident within a year.

Step-by-step Solution (Exactly as in slides)

Let:

- A_1 = event that policyholder has an accident within a year
- A = event that policyholder is accident prone

Then:

$$Pr(A_1) = Pr(A_1 \mid A)Pr(A) + Pr(A_1 \mid A^c)Pr(A^c)$$

Substitute values:

$$Pr(A_1) = (0.4)(0.3) + (0.2)(0.7)$$

$$Pr(A_1) = 0.12 + 0.14 = 0.26$$

✅ Final Answer:

$$Pr(A_1) = 0.26$$

1.3 Bayes Formula — Example #3.1 (Part 2/2)

Problem Statement (Twist)

Suppose the new policyholder **has** an accident within a year.
Find:

$$Pr(A \mid A_1)$$

Step-by-step Solution (Exactly as in slides)

$$Pr(A \mid A_1) = \frac{Pr(AA_1)}{Pr(A_1)}$$

Now:

$$Pr(AA_1) = Pr(A)\,Pr(A_1 \mid A)$$

So:

$$Pr(A \mid A_1) = \frac{Pr(A)Pr(A_1 \mid A)}{Pr(A_1)}$$

Substitute values:

$$Pr(A \mid A_1) = \frac{(0.3)(0.4)}{0.26}$$
$$= \frac{0.12}{0.26} = \frac{6}{13}$$

✅ Final Answer:

$$Pr(A \mid A_1) = \frac{6}{13}$$

1.4 Formal Introduction — Law of Total Probability (Formula 3.4)

Assumptions (Must be explicitly stated)

Suppose:

- $B_1, B_2, \dots, B_n$ are **mutually exclusive events**
- Their union forms the whole event space B:

$$\bigcup_{i=1}^{n} B_i = B$$

Meaning:

> Exactly one of the events $B_1, B_2, \dots, B_n$ must occur.

Derivation (As shown in slides)

Write:

$$A = \bigcup_{i=1}^{n} AB_i$$

Since the events AB_i are mutually exclusive:

$$Pr(A) = \sum_{i=1}^{n} Pr(AB_i)$$

Convert using conditional probability:

$$Pr(AB_i) = Pr(A \mid B_i)Pr(B_i)$$

Thus:

$$Pr(A) = \sum_{i=1}^{n} Pr(A \mid B_i)Pr(B_i)$$

✅ This is the **Law of Total Probability (Formula 3.4)**.

1.5 Bayes Formula (Proposition 3.1)

Key Identity Used

The slides state:

$$Pr(AB_i) = Pr(B_i \mid A)Pr(A)$$

Bayes Formula (Final Result from slides)

$$Pr(B_i \mid A) = \frac{Pr(A \mid B_i)Pr(B_i)}{\sum_{j=1}^{n} Pr(A \mid B_j)Pr(B_j)}$$

This is known as the **Bayes Formula (Proposition 3.1)**.

Definitions (Explicit from slides)

- $Pr(B_i)$ = **apriori probability**

  > probabilities that are formed from self-evident or presupposed models

- $Pr(B_i \mid A)$ = **posteriori probability**

  > probabilities that are derived or calculated after observing certain events of event B_i given A

1.6 Learning by Example — Example #3.2 (Cards)

Problem Statement

There are 3 cards:

1. RR (both sides red)
2. BB (both sides black)
3. RB (one side red, one side black)

Cards are mixed, one card selected and placed down.
If the upper side is **red**, find probability that the other side is **black**.

Step-by-step Solution (Bayes Method from slides)

Let:

- RR, BB, RB denote events that chosen card is those types
- R = event that the upturned side is red

We want:

$$Pr(RB \mid R)$$

Bayes Setup

$$Pr(RB \mid R) = \frac{Pr(RB \cap R)}{Pr(R)}$$

Convert numerator:

$$Pr(RB \cap R) = Pr(R \mid RB)Pr(RB)$$

Denominator expands using total probability over card types:

$$Pr(R) = Pr(R \mid RR)Pr(RR) + Pr(R \mid RB)Pr(RB) + Pr(R \mid BB)Pr(BB)$$

So:

$$Pr(RB \mid R) = \frac{Pr(R \mid RB)Pr(RB)}{Pr(R \mid RR)Pr(RR) + Pr(R \mid RB)Pr(RB) + Pr(R \mid BB)Pr(BB)}$$

Substitute values from the slide

- $Pr(RB) = Pr(RR) = Pr(BB) = \frac{1}{3}$
- $Pr(R \mid RR) = 1$
- $Pr(R \mid RB) = \frac{1}{2}$
- $Pr(R \mid BB) = 0$

So:

$$Pr(RB \mid R) = \frac{(1/2)(1/3)}{(1)(1/3) + (1/2)(1/3) + (0)(1/3)}$$

Compute:

Numerator:

$$\frac{1}{6}$$

Denominator:

$$\frac{1}{3} + \frac{1}{6} + 0 = \frac{1}{2}$$

Thus:

$$Pr(RB \mid R) = \frac{1/6}{1/2} = \frac{1}{3}$$

✅ Final Answer:

$$Pr(RB \mid R) = \frac{1}{3}$$

Slide Explanation of Common Mistake

Many guess $1/2$ incorrectly by assuming two equally likely cases (RR or RB).
The slides explain the correct reasoning by listing 6 equally likely sides:

$$R_1, R_2, B_1, B_2, R_3, B_3$$

Given a red side, only outcomes are R_1, R_2, R_3.
Only R_3 corresponds to RB card having black on the other side.
So probability = $1/3$.

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2) Random Variables

2.1 Motivation and Concept

Key Motivation (From slides)

When an experiment is performed, we are frequently interested mainly in some **function** of the outcome rather than the actual outcome itself.

Examples from slides:

- Dice tossing: focus on **sum** of dice rather than individual values.
- Coin flipping: interest may lie in **total number of heads** rather than exact sequence.

Definition (From slide)

These real-valued functions defined on the sample space are called:

> **Random Variables**.

Properties (explicit from slide):

- Values are determined by outcomes of an experiment.
- Probabilities are assigned to possible values of Random Variables.

Formal Definition (From slide)

A random variable X on a sample space Ω is a function:

$$X : \Omega \to \mathrm{R}$$

It assigns to each sample point $\omega \in \Omega$ a real number:

$$X(\omega)$$

Discrete Restriction (Assumption stated in slides)

Until further notice, attention is restricted to **discrete** random variables, meaning:

- values are in a range that is **finite** or **countably infinite**
- even though X maps into R, the set of values X takes is a **discrete subset** of R.

2.2 Distribution of a Random Variable

What the slides compare

Probability space (earlier lecture) defines:

1. sample space Ω
2. probability of each sample point

Analogously, random variable defines:

1. set of values it can take
2. probabilities with which it takes those values

Event Form

Let a be a number in the range of X. Then:

$$\{\omega \in \Omega : X(\omega) = a\}$$

This is an event in the sample space.
Abbreviated as:

$$X = a$$

Then:

$$Pr[X = a]$$

The collection of these probabilities for all values of a is called the **distribution** of X.

2.3 Visualization (Bar Diagram)

The distribution can be visualized as a bar diagram:

- x-axis: values random variable can take
- bar height at value a: $Pr[X = a]$
- each probability computed by probability of corresponding event in sample space

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2.4 Random Variable Example (Example-1)

Problem

Experiment: toss 3 fair coins.
Let Y = number of heads.

Possible values:

$$Y \in \{0, 1, 2, 3\}$$

Step-by-step Probabilities (From slides)

$$P\{Y = 0\} = P\{(t, t, t)\} = \frac{1}{8}$$

$$P\{Y = 1\} = P\{(t, t, h), (t, h, t), (h, t, t)\} = \frac{3}{8}$$

$$P\{Y = 2\} = P\{(t, h, h), (h, t, h), (h, h, t)\} = \frac{3}{8}$$

$$P\{Y = 3\} = P\{(h, h, h)\} = \frac{1}{8}$$

Total Probability Check (From slides)

Since Y must take one of the values 0 through 3:

$$1 = P\left(\bigcup_{i=0}^{3}\{Y = i\}\right) = \sum_{i=0}^{3} P(Y = i)$$

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3) Probability Mass Function (PMF)

3.1 Discrete Random Variable Definition

A random variable that can take on at most a **countable number** of possible values is said to be **discrete**.

3.2 PMF Definition (Formal)

Let X be a discrete random variable with range:

$$R_X = x_1, x_2, x_3, \ldots$$

(finite or countably infinite)

The function:

$$p(x_k) = Pr(X = x_k)$$

is called the **Probability Mass Function (PMF)** of X.

3.3 PMF Property (Normalization)

Since X must take on one of the values x_k, we have:

$$\sum_k p(x_k) = 1$$

3.4 PMF Example (Given in slides)

Problem

A random variable X has PMF:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

where λ is positive.

Find:

1. $P\{X = 0\}$
2. $P\{X > 2\}$

Step 1: Use the PMF Normalization Condition

Since:

$$\sum_{i=0}^{\infty} p(i) = 1$$

Substitute:

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Step 2: Use Series Identity from slide

The slide states:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

So:

$$ce^{\lambda} = 1$$

Thus:

$$c = e^{-\lambda}$$

Step 3: Compute $P\{X = 0\}$

$$P\{X = 0\} = p(0) = c\frac{\lambda^0}{0!}$$

Substitute:

- $c = e^{-\lambda}$
- $\lambda^0 = 1$
- $0! = 1$

$$P\{X = 0\} = e^{-\lambda}$$

✅ Final Answer:

$$P\{X = 0\} = e^{-\lambda}$$

Step 4: Compute $P\{X > 2\}$

Slides use complement:

$$P\{X > 2\} = 1 - P\{X \leqslant 2\}$$

Then:

$$P\{X > 2\} = 1 - P\{X = 0\} - P\{X = 1\} - P\{X = 2\}$$

Final Exam Revision Checklist (From Lecture 5)

Bayes & Total Probability

You must know:

- $A = AB \cup AB^c$
- $Pr(A) = Pr(A \mid B)Pr(B) + Pr(A \mid B^c)[1 - Pr(B)]$
- Law of Total Probability:

$$Pr(A) = \sum_{i=1}^{n} Pr(A \mid B_i)Pr(B_i)$$

- Bayes Formula:

$$Pr(B_i \mid A) = \frac{Pr(A \mid B_i)Pr(B_i)}{\sum_{j=1}^{n} Pr(A \mid B_j)Pr(B_j)}$$

Random Variables

You must know:

- $X : \Omega \to \mathrm{R}$
- distribution = all $Pr[X = a]$

PMF

You must know:

- discrete RV = countable values
- $p(x_k) = Pr(X = x_k)$
- $\sum_k p(x_k) = 1$

If you want, I can now also make this into a **super clean "exam notebook style" PDF-format writeup** (same content, just formatted like assignment notes) 📄✨